



Outdoor Field Testing

Range Test Report

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Preface

This document presents the findings and analysis of the outdoor field-testing range conducted to evaluate the signal range and performance of the tested system in an outdoor environment.

Privacy Information

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1.Introduction

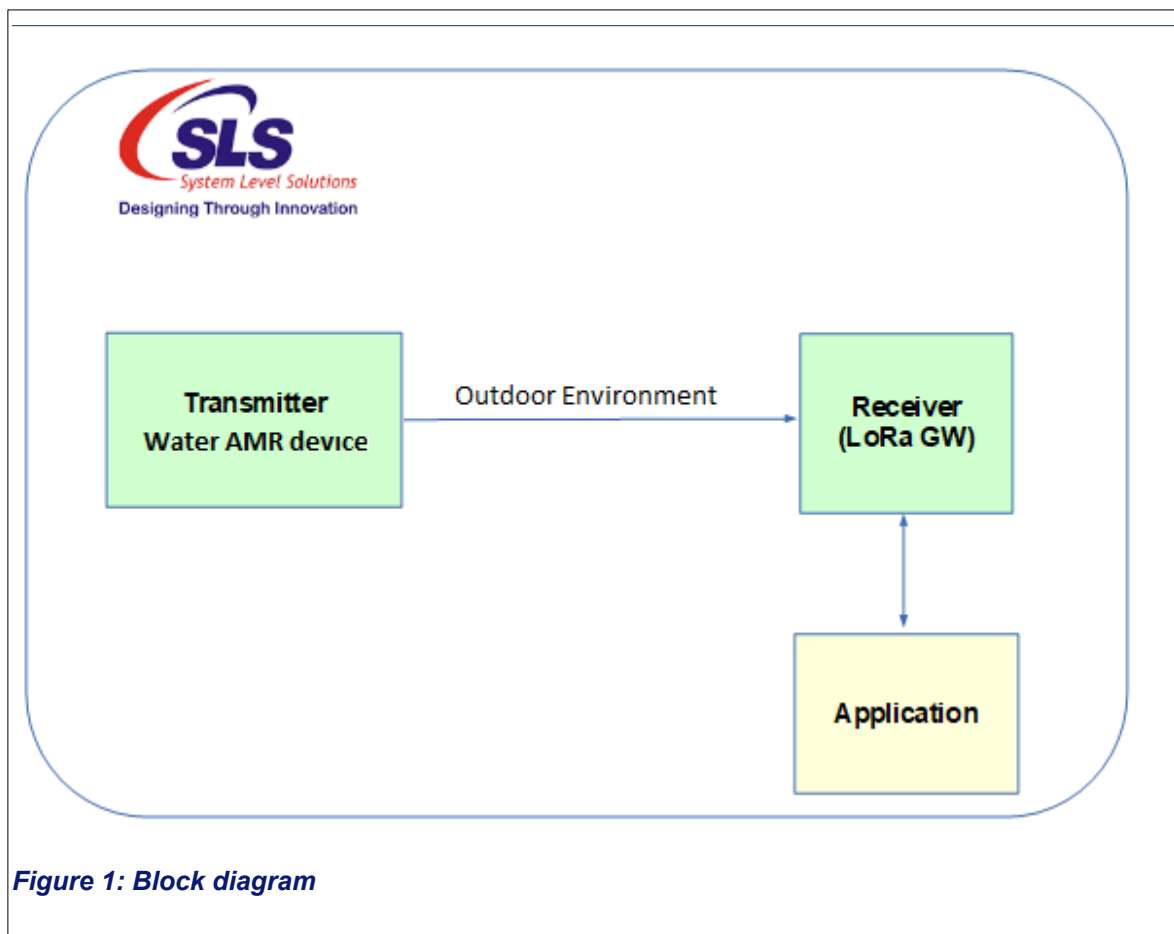
- This document describes the information about Out-Door RF Range Testing of Water AMR devices at Mohanpura (Roorkee).
- When selecting right Radio Technology, the achievable communication range is an important factor.

1.1 Why Range Test Performed?

- Range tests are performed to determine the maximum distance or extent over which a particular device, system, or object can operate effectively. They help evaluate the performance, reliability, and limitations of the subject under consideration within a specific range of operation.

2. Block diagram

- The block diagram represents a visual depiction of the setup for transmitting and receiving signals in an urban environment. It provides a graphical representation of the components and connections involved in the transmission and reception process, allowing for a better understanding of the system's configuration and operation.



- As above block diagram shows graphical representation of Transmitter and Receiver setup in Client site location.

3. Range Testing with Standard Proven Handy Tool

- **RF Power Meters:** These tools measure the power level of RF signals, allowing you to assess signal strength and determine the range of a transmitter.
- **Spectrum Analyzers:** Spectrum analyzers provide a visual representation of the frequency spectrum, enabling you to identify and analyze RF signals present in the environment. They help in detecting interference and optimizing frequency usage.
 - **Noise Level measurement:** The noise level in an RF survey refers to the amount of unwanted electromagnetic signals or interference present in the radio frequency environment. It is an important factor to consider when evaluating the quality and reliability of wireless communication systems, as high levels of noise can degrade signal clarity and reduce the overall performance of RF devices.
- **Field Strength Meters:** These devices measure the strength of RF signals at specific locations, allowing you to map out signal coverage and identify areas with weak or strong signal reception.
- **Signal Generators:** Signal generators produce RF signals of known frequency and power, which can be used to test the sensitivity and range of receivers.

4.Site location

- **Site Location:** Mohanpura,Roorkee
- **Lat/Long data:** 29.845808 N,77.8897058 E



Figure 2: Mohanpura, Roorkee Client site

5. Water AMR device



LoRa Module Device Configuration:

- Module: NLN500A
- Output Power: +21 dBm
- Operating Frequency: 865MHz to 868MHz
- Spreading Factor: SF9
- Coding Rate: 4/8
- Transmit Time: 1 minute

* Range test is performed with power level of +21 dBm.

6.NLG7644 (LoRa Gateway)

- Gateway location optimization refers to the process of strategically determining the optimal placement of gateways in a network infrastructure. This involves selecting suitable locations that maximize network coverage, minimize signal interference, and ensure efficient communication between devices and the gateway. By optimizing gateway locations, network performance can be enhanced, ensuring reliable connectivity and effective data transmission throughout the network.



Figure 4: NLG7644 LoRa Gateway 1 at Mohanpura (Roorkee)

7.Site analysis

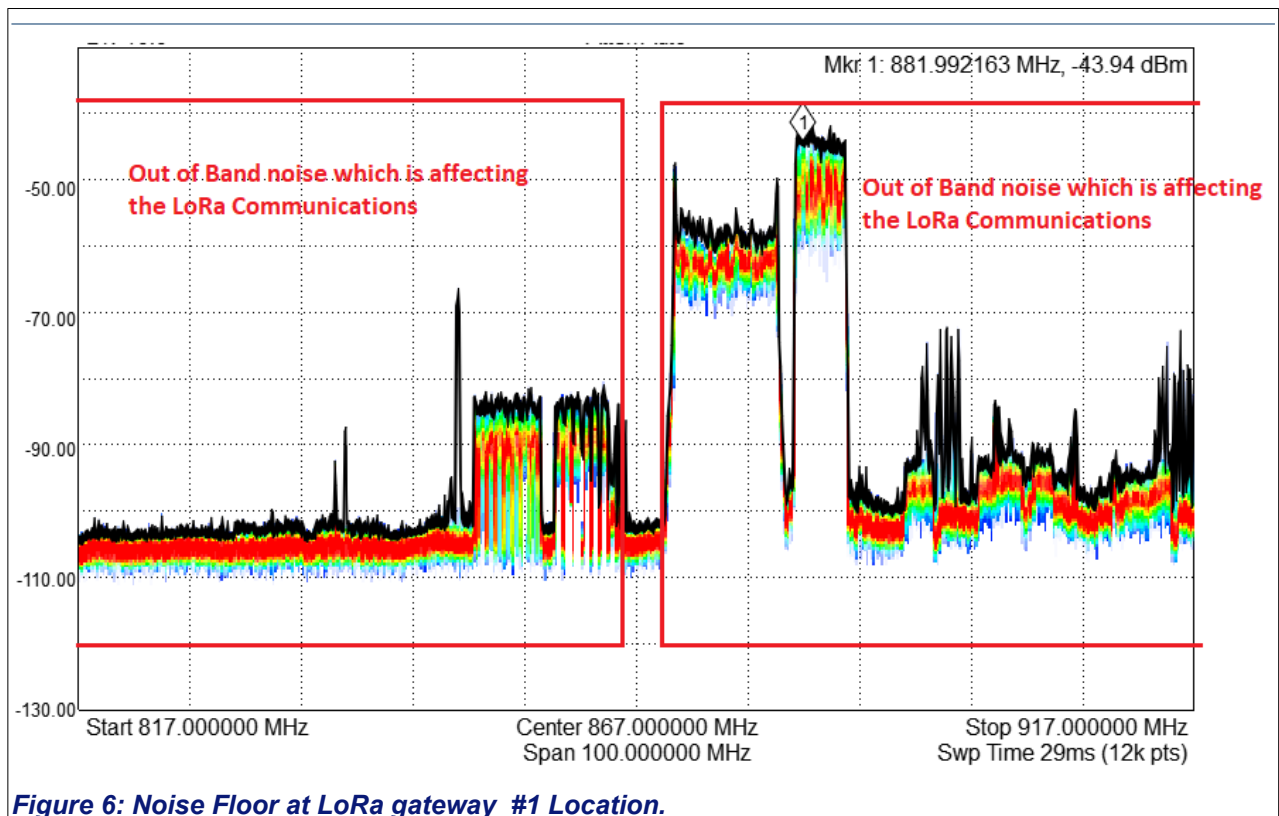
- During the Mohanpura (Roorkee) RF site survey, Client had installed around 1300 water AMR meters with one NLG7644 (LoRa Gateway).
- Each house having a separate water supply pipeline with one smart AMR meter was installed inside or outside the house area.
- In some house water meter was open towards sky and maintain the 3 feet height from the ground which gave the good connectivity.
- But other houses water meters were surrounded or inside House and closed the house entrance by metal doors (Figure 5).
- There are more than one GSM towers near the buildings which continuously producing the out-band noise.

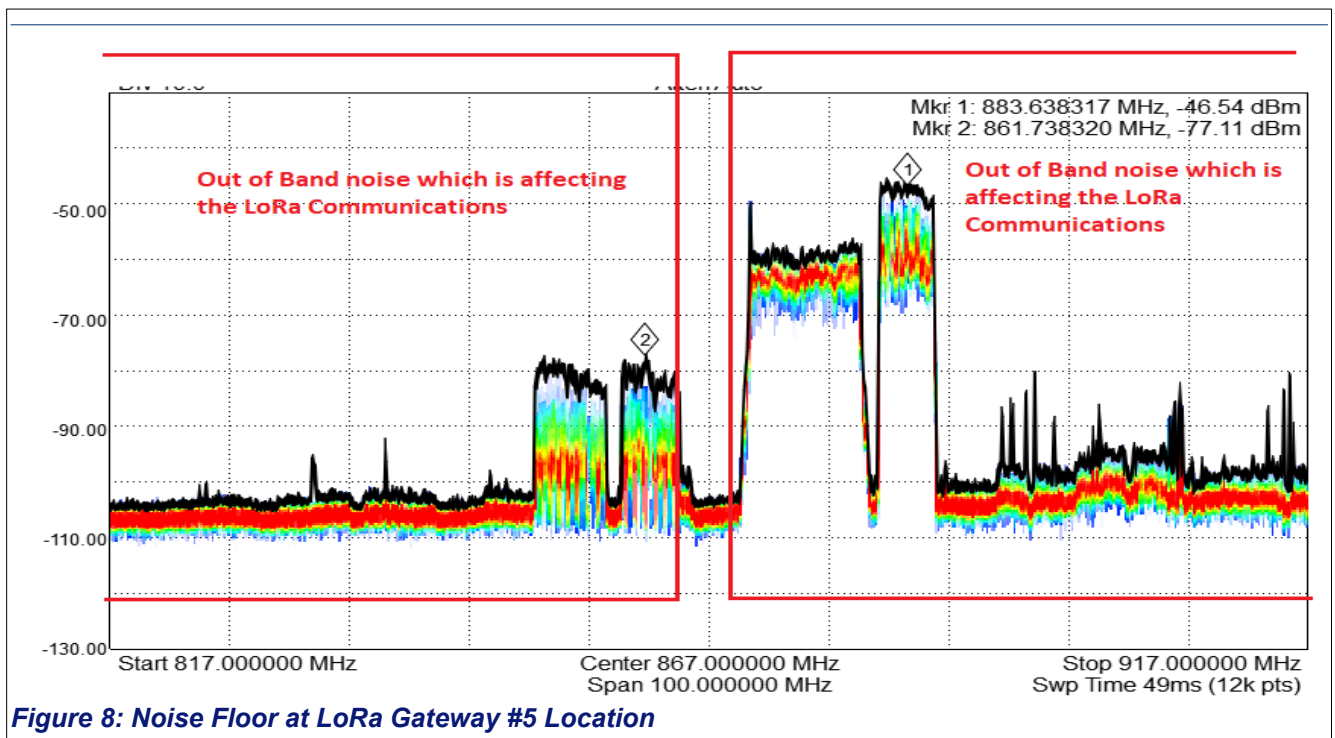
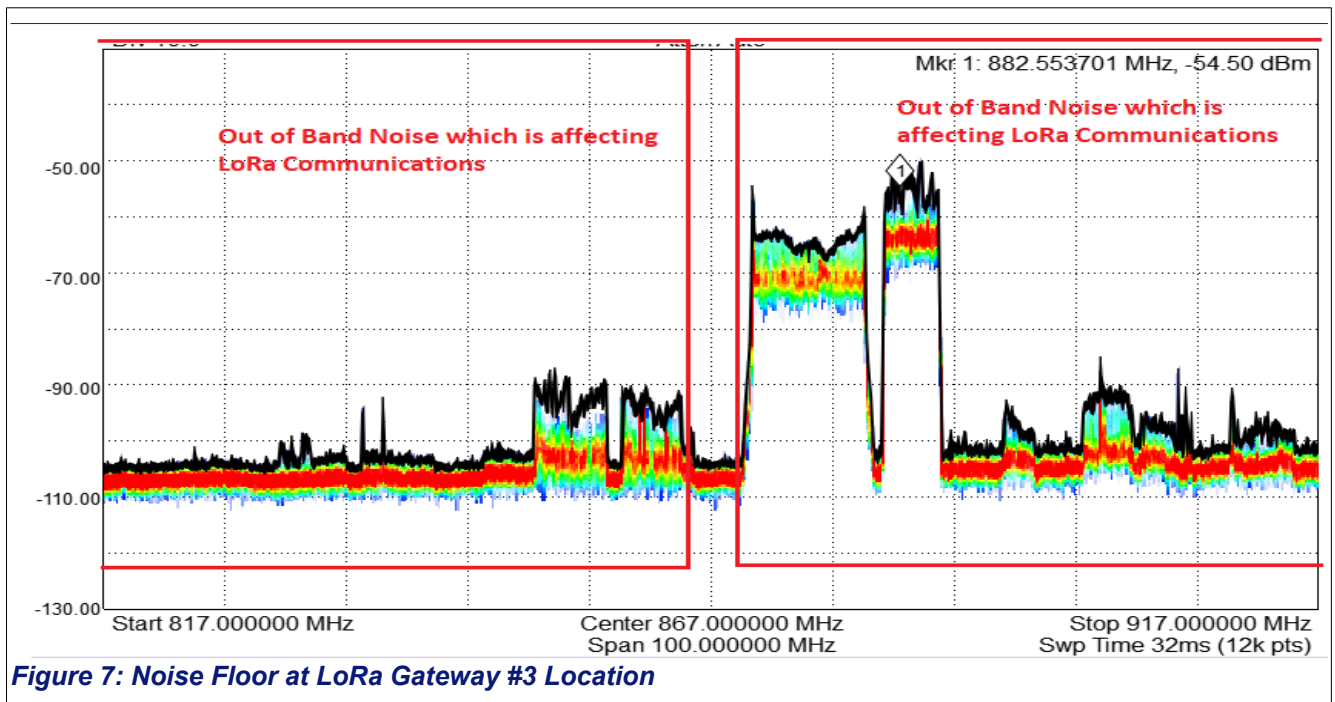


Figure 5: Improper Placement of Devices(Inside House and closed by Metal Door)

8. Test Procedure

- To investigate the presence of interference in the LoRaWAN networks, we conducted a comprehensive interference-hunting campaign using specialized equipment such as spectrum analyzer, Avartanam filter & Signal generator.
- We collected empirical data by taking measurements at various locations and frequencies and analyzed the data to identify the sources of interference.
- We also identified 5 different places for LoRa gateway installation to cover the entire 1300 water AMR devices on the Mohanpura (Roorkee) Client site and which has been selected after by observing the noise and installation parameters.





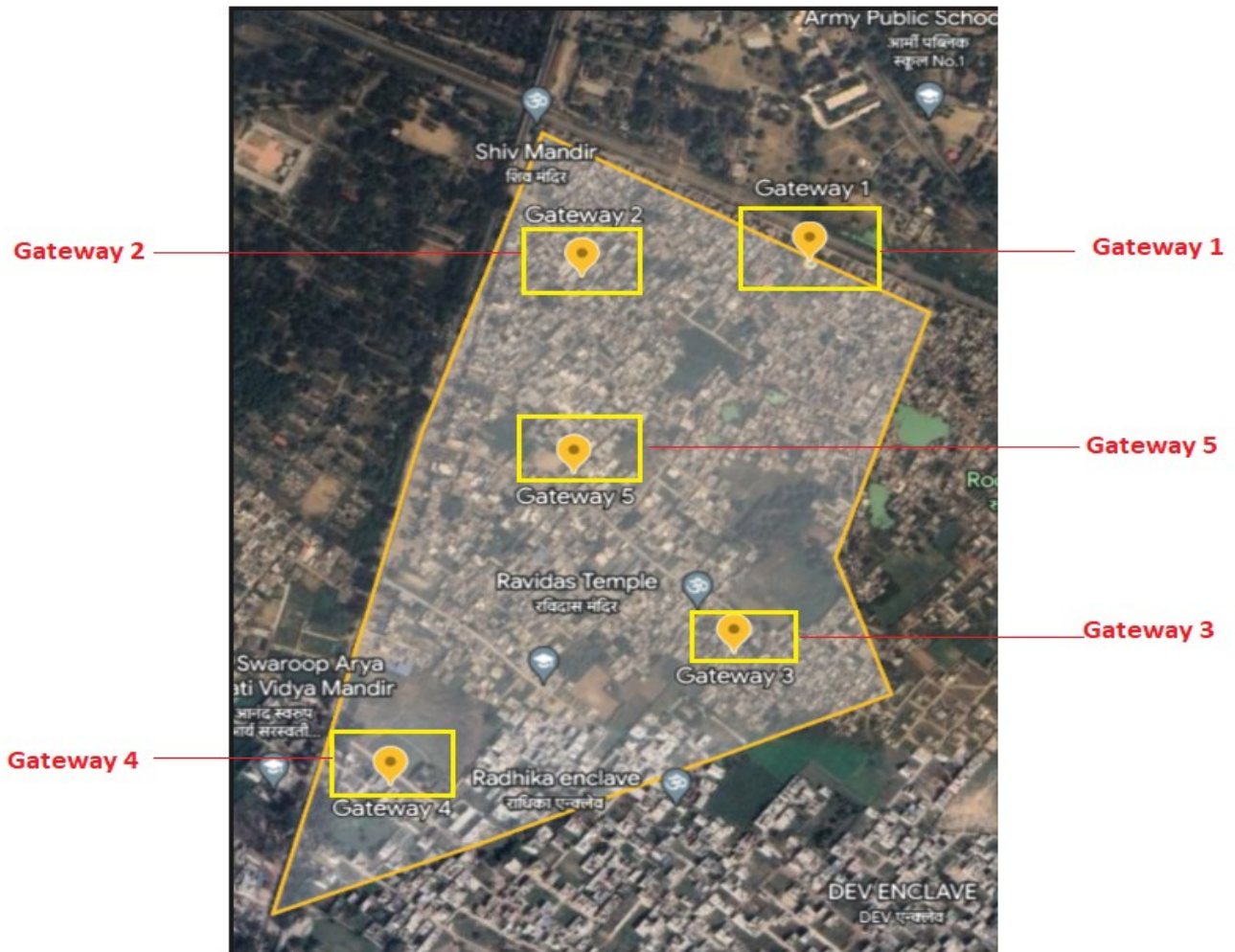


Figure 9: LoRa Gateway Locations.

- During the RF site survey, we placed the LoRa gateway #1 at mentioned location (Figure 11) to cover test points (A to F) water AMR devices locations.
- Afterward our team placed the Water AMR device in randomly on the Different location on client site near to Gateway #1 locations and some was tested on the open-air environments, and some devices tested with already installed water AMR devices.

- After the RF site survey analysis, we have estimated that client need to install the total 5 gateway to cover the entire 1300 Water AMR devices as shown in figure 9.

Gateway No.	Latitude	Longitude
1	29.846669 N	77.890716 E
2	29.844218 N	77.892997 E
3	29.839575 N	77.891257 E
4	29.841188 N	77.889022 E
5	29.843716 N	77.888081 E

Table 1: Lat/Long Locations Data of all LoRa Gateway

- For LoRa Gateway #2 and #4 client faced the permission delay related issue during the testing time and so we haven't carried out testing points for LoRa Gateway #2 and #4.
- The AMR device is running on the battery supply, on every 1 min device sent the uplink message and checked the RSSI value on the AMR debug portal/website for each location.
- Our team also carried out range test with the LoRa Gateway #3 and #5 at different location at Mohnapura site inside and outside the residence house.

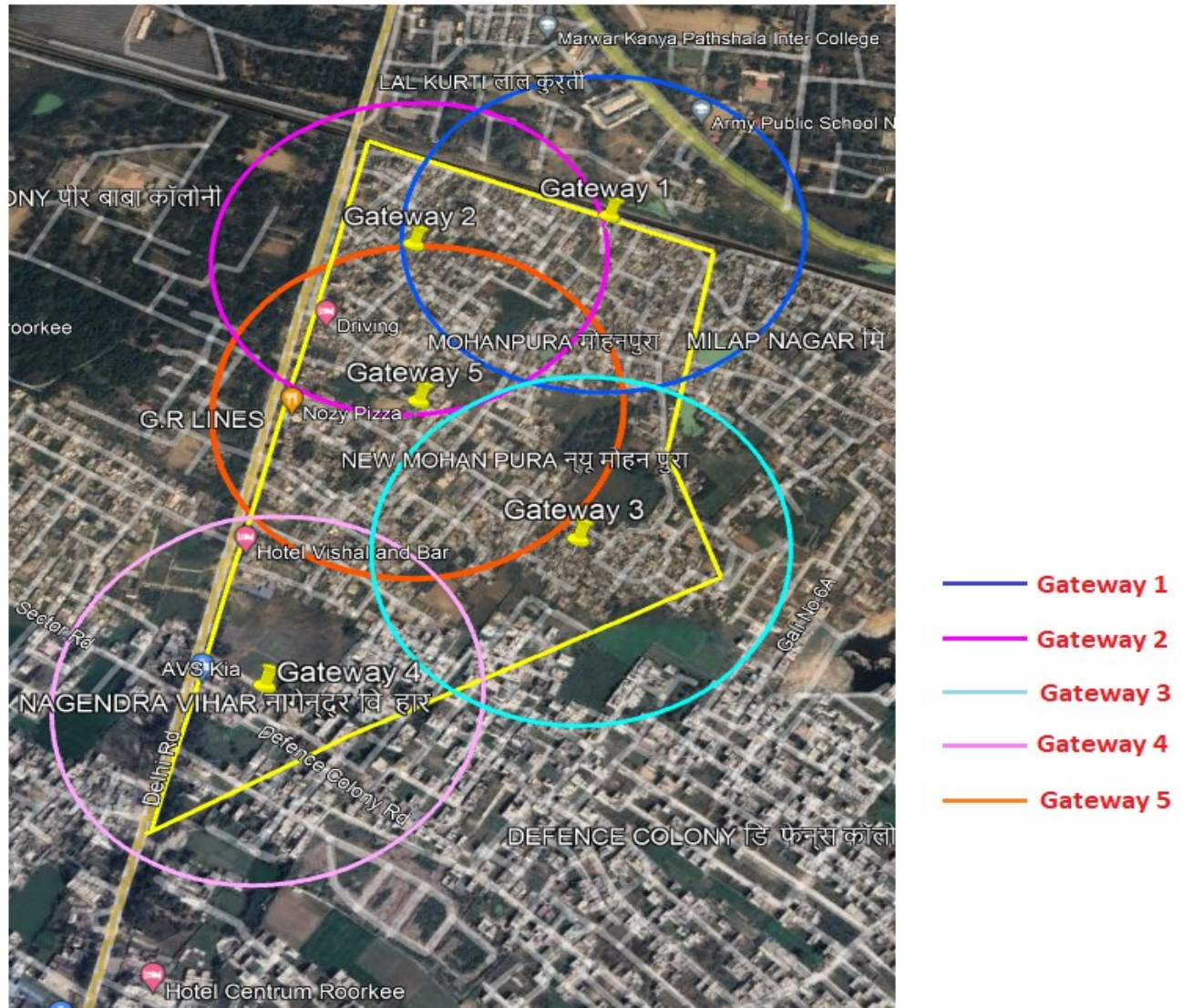


Figure 10: LoRa Gateway Range Circle

9.Measurement Results

- LoRa Gateway #1 Test Points:

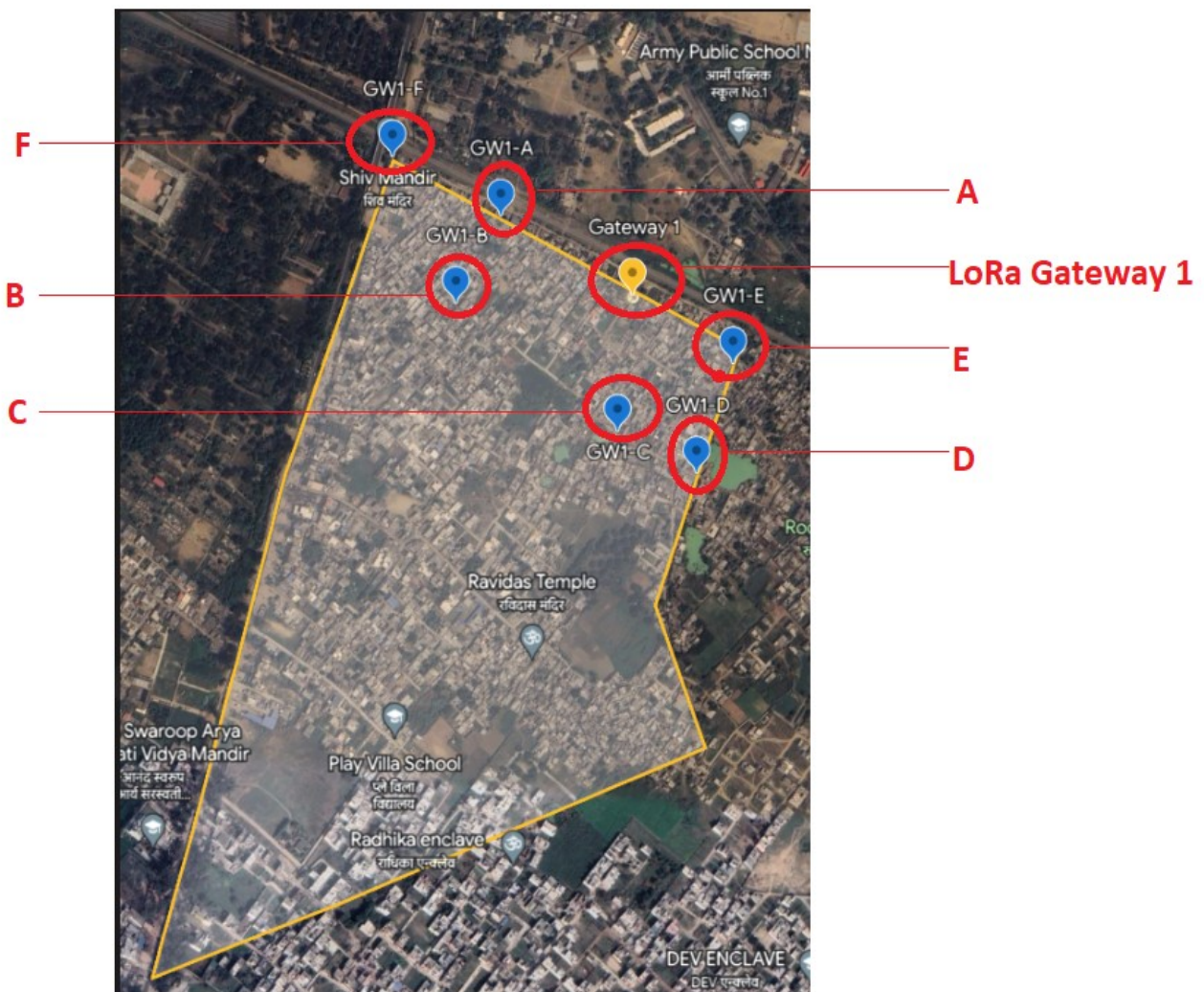


Figure 11: LoRa Gateway #1 Test Points

- LoRa Gateway #1 Test Results:

Gateway Location: NLG7644 LoRa Gateway 1 (26.846669,77.890716)		
Node Location: Test Point A		
Data Rate	RSSI	Packet success ratio (%)
SF9	-92.5	100
	-85.8	
	-87.2	
Node Location: Test Point B		
Data Rate	RSSI	Packet success ratio (%)
SF9	-92	100
	-90.8	
	-87.5	
Node Location: Test Point C		
Data Rate	RSSI	Packet success ratio (%)
SF9	-95	100
	-95.5	
	-92.5	
Node Location: Test Point D		
Data Rate	RSSI	Packet success ratio (%)
SF9	-95.5	100
	-98.2	
	-95.2	
Node Location: Test Point E		
Data Rate	RSSI	Packet success ratio (%)
SF9	-93	100
	-96	
	-93.8	
Node Location: Test Point F		
Data Rate	RSSI	Packet success ratio (%)
SF9	-	Not connected

Table 2: LoRa Gateway #1 Test Results.

- There will be variation in the RSSI value due to the bad weather condition or any object passing between the AMR & Gateway path.

- **LoRa Gateway #5 Test Points:**

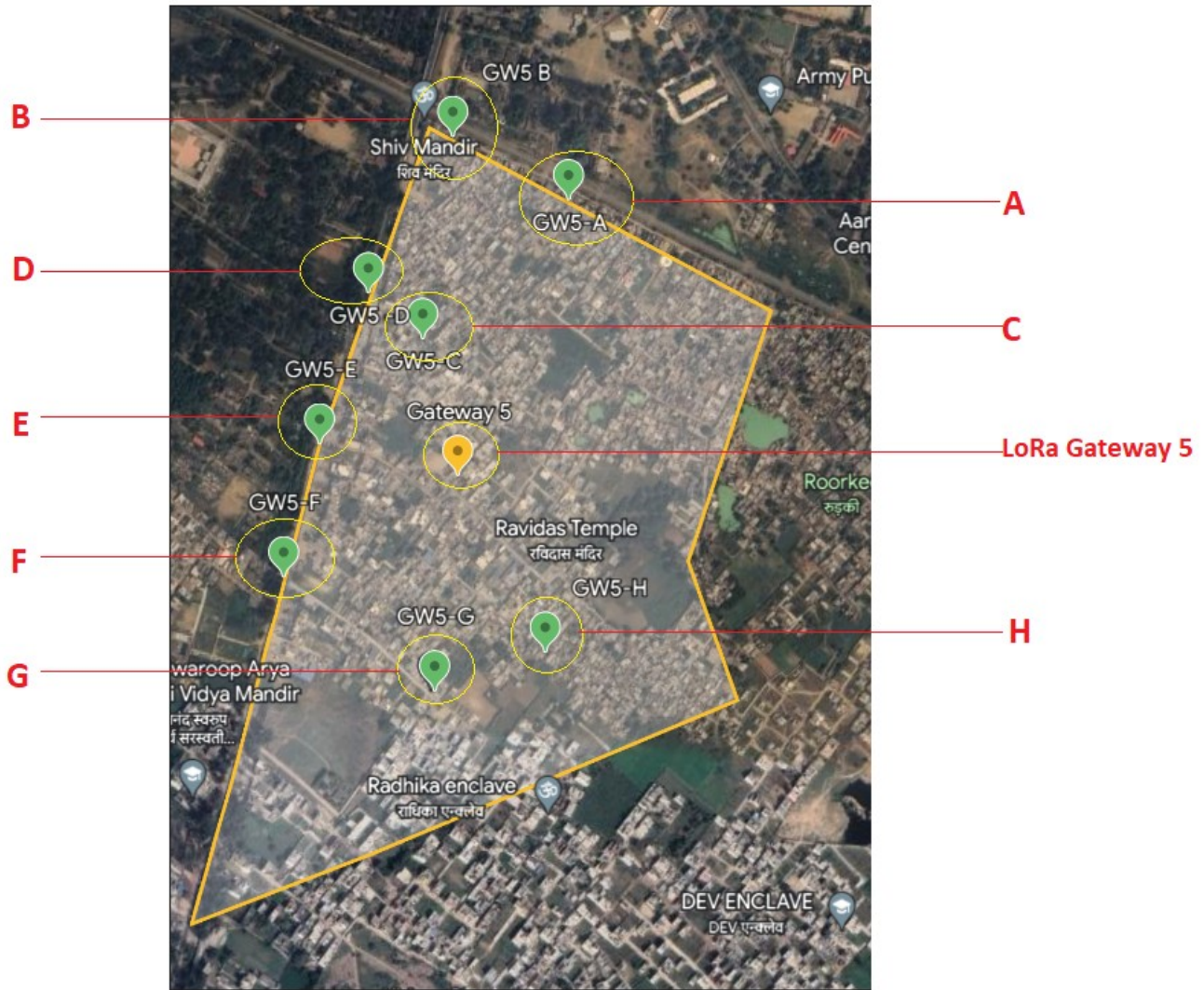


Figure 12: LoRa Gateway #5 Test Points

- LoRa Gateway #5 Test Results:

Gateway Location: NLG7644 LoRa Gateway 5 (29.843716,77.888081)		
Node Location: Test Point A		
Data Rate	RSSI	Packet success ratio (%)
SF9	-	Not connected
Node Location: Test Point B		
Data Rate	RSSI	Packet success ratio (%)
SF9	-	Not connected
Node Location: Test Point C		
Data Rate	RSSI	Packet success ratio (%)
SF9	-94.2	100
	-100.8	
Node Location: Test Point D		
Data Rate	RSSI	Packet success ratio (%)
SF9	-104.2	100
	-99.2	
Node Location: Test Point E		
Data Rate	RSSI	Packet success ratio (%)
SF9	-99	100
	-94.5	
Node Location: Test Point F		
Data Rate	RSSI	Packet success ratio (%)
SF9	-89.8	100
	-92.5	
Node Location: Test Point G		
Data Rate	RSSI	Packet success ratio (%)
SF9	-106.8	100
	-106.5	
Node Location: Test Point H		
Data Rate	RSSI	Packet success ratio (%)
SF9	-94	100
	-89.8	

Table 3: LoRa Gateway #5 Test Results.

- There will be variation in the RSSI value due to the bad weather condition or any object passing between the AMR & Gateway path.

10. Suggestions

Our team has thoroughly studied a water meter location at Mohanpura (Roorkee) client site and carefully understood the requirement. The LoRa Gateway required sizeable space that would be safe and secure.

- Increase height of the Water AMR device at least 3 feet above from the ground level for better RF Range performance.
- From the noise floor data all the gateways must require to be installed with Avartanam filter on mentioned location in Figure 9.
- During the testing test point F Water AMR devices was not connected with the Gateway #1. So, for that Client must have to install LoRa gateway #2 on the suggested locations to connect the test point F water AMR devices.
- Also, Client need to increase the height of LoRa Gateway #3 and #5 to improve the RF range like LoRa Gateway #1.
- We require all installed water AMR devices Lat/Long data for proper identified the blind spot devices area information for better RF survey.
- From that Lat/long data, we clearly get the idea about the which devices are connected or not connected (blind spot area) with the nearest gateway.
- Battery with Super-Cap is required due to alone battery is not delivered the enough power (+21 dBm).
- Our team have also cross verified performance of both cases with and without Super-Cap. With super-cap AMR devices successfully connected with the LoRa gateway during the testing time.
- Battery with super-cap will also improve the RF range performance.
- Put Lightning Arrester with the LoRa gateway for Preventing Product Damage against over-voltage and surge current induced by bolts of lightning.